

DIAMOND SHOALS BUOY 41025, USA

WMO: 997040

Lat: 35.025N

Lon: 75.363W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.4	34.5	13.8	11.1	33.4	17.9	13.6	37.2	75.9	56.6	64.9	55.6	26.3	340	0.744

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	85.3	78.5	83.7	77.4	82.9	76.8	80.9	84.7	79.8	83.0	78.6	81.9	13.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.9	155.4	84.1	78.7	149.0	82.7	77.6	143.5	81.6	44.5	84.4	43.2	82.9	42.1	81.8	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
58.9	48.9	40.2	27.5	85.4	5.9	1.7	23.3	86.6	19.9	87.6	16.6	88.5	12.3	89.7		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.8	51.7	52.3	55.7	61.7	68.1	76.1	79.4	79.2	76.4	69.8	62.1	56.8
	DBStd	12.02	10.24	8.53	8.15	6.55	5.20	4.32	2.56	2.60	3.13	6.10	7.80	9.15
	HDD50	270	107	69	39	3	0	0	0	0	0	0	9	43
	HDD65	1706	423	362	301	136	32	2	0	0	0	28	143	279
	CDD50	6054	159	133	215	355	561	783	912	905	793	613	371	254
	CDD65	2015	10	6	12	38	128	335	447	440	343	176	55	25
	CDH74	13696	2	0	0	7	146	2259	4205	4094	2221	667	66	29
	CDH80	1739	0	0	0	0	1	239	665	668	141	25	0	0
Wind	WSAvg	19.6	22.8	23.9	23.9	22.6	19.6	15.9	16.7	14.4	17.1	17.2	19.9	21.5
Precipitation	PrecAvg	50.9	4.0	3.4	3.8	3.2	3.7	4.7	5.5	6.2	5.6	4.3	3.1	3.6
	PrecMax	71.2	8.0	6.3	7.2	7.8	8.9	7.6	9.9	10.4	10.6	8.9	8.0	6.7
	PrecMin	28.9	1.4	0.9	1.1	0.9	0.1	0.8	1.5	2.6	0.7	0.4	0.6	1.3
	PrecStd	8.0	1.8	1.5	1.6	1.8	2.3	1.9	1.9	2.3	2.7	2.3	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.0	72.7	73.3	75.0	78.1	91.4	91.7	85.6	83.2	82.1	77.5	77.1
		MCWB	69.0	67.9	67.0	68.0	70.5	82.3	81.8	79.7	75.6	76.2	72.2	73.0
	2%	DB	71.5	71.0	71.8	73.4	76.8	84.0	86.1	84.5	82.3	80.6	75.9	74.6
		MCWB	65.5	66.3	64.8	66.9	70.5	78.3	77.5	78.4	75.3	74.8	69.7	69.4
	5%	DB	70.0	68.3	70.0	72.3	75.6	82.1	83.5	83.8	81.6	79.6	74.5	72.2
		MCWB	63.9	62.8	63.3	65.9	69.6	77.0	77.5	77.4	75.2	73.7	67.9	66.3
	10%	DB	67.7	65.9	68.2	70.9	74.7	80.8	82.6	83.0	80.9	77.9	73.0	70.2
		MCWB	61.9	60.3	61.7	64.8	68.9	75.9	77.1	76.6	74.6	71.0	66.3	64.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.6	69.3	69.1	71.0	74.4	82.8	83.2	81.7	78.7	78.1	73.7	73.8
		MCDB	73.5	72.1	72.6	73.8	76.2	91.1	90.4	84.1	81.1	81.1	76.8	77.0
	2%	WB	66.7	67.6	66.3	68.9	72.9	80.1	81.0	80.5	77.8	76.2	71.6	70.9
		MCDB	70.7	71.4	70.4	72.2	75.2	84.7	84.7	83.6	80.8	80.0	75.0	74.4
	5%	WB	64.6	64.3	64.6	67.2	71.7	78.3	79.9	79.4	76.9	74.5	69.6	67.5
		MCDB	69.5	67.9	69.1	71.1	74.3	81.4	82.8	83.2	80.4	79.0	74.0	71.9
	10%	WB	61.9	60.9	62.2	65.9	70.5	76.7	78.9	77.9	75.9	73.0	67.5	65.1
		MCDB	67.0	65.6	67.2	70.2	73.8	79.9	81.8	82.1	79.9	78.0	72.6	69.9
Mean Daily Temperature Range	5% DB	MDBR	11.0	10.9	10.2	7.9	6.2	4.8	4.7	4.0	4.3	5.8	8.2	10.0
		MCDBR	11.3	12.9	11.4	8.3	6.1	6.6	6.3	4.3	4.6	5.9	9.0	11.2
		MCWBR	12.0	13.3	10.8	8.6	5.9	5.3	5.1	4.1	4.5	5.9	9.2	10.9
	5% WB	MCDBR	11.7	13.0	11.8	8.7	5.4	7.0	6.2	3.7	4.5	6.5	9.7	11.7
		MCWBR	12.4	13.7	11.8	9.4	6.4	5.7	5.4	4.1	4.6	7.1	10.9	12.3
Clear-Sky Solar Irradiance	taub	0.341	0.345	0.366	0.407	0.447	0.480	0.503	0.491	0.425	0.376	0.351	0.346	
	taud	2.513	2.502	2.459	2.347	2.274	2.229	2.179	2.211	2.396	2.506	2.527	2.514	
	Ebn at Noon	273	285	287	279	268	258	251	252	265	271	267	264	
	Edh at Noon	26	30	34	39	43	45	47	44	36	29	26	25	
All-Sky Solar Radiation	RadAvg	733	982	1322	1703	1943	2024	1882	1739	1440	1185	841	674	
	RadStd	53	106	111	137	138	106	122	116	97	108	63	54	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+313	+211

Nomenclature: See separate page